

Sayan Patra

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EDUCATION

Maulana Abul Kalam Azad University of Technology (WBUT)
B.Tech in Computer Science and Engineering | **CGPA: 7.63**

Aug 2022 – Present
Kolkata, West Bengal

Midnapore Collegiate School
Higher Secondary (WBCHSE) | **87.4%**

2019 – 2021
West Bengal

EXPERIENCE

Founder & Full Stack Developer
July 2024 – Present

Messmate 

- Built and scaled a MERN-based platform connecting PG owners with students, enabling real-time booking and availability tracking.
- Designed and developed 40+ scalable REST APIs for authentication, booking workflows, and notifications.
- Engineered a real-time booking system using Socket.io, ensuring data consistency and low-latency updates.
- Architected a modular backend system optimized for concurrent users and high-performance query execution.
- Implemented secure authentication using JWT and HTTP-only cookies.
- Optimized MongoDB queries, improving response time by 40%.
- Integrated geolocation-based search using OpenStreetMap for location-aware discovery.
- Deployed production applications on Vercel and Render with CI/CD pipelines for automated builds and deployments.

PROJECTS

E-commerce Platform (BlackPanda)  | *Node.js, Express.js, MongoDB*

- Built a full-stack e-commerce platform with authentication, product management, and cart functionality.
- Designed RESTful APIs and implemented efficient CRUD operations for scalable data handling.

AI Interview Feedback System (LLM + RAG Based) (Ongoing) | *React, Next.js, Python, Machine Learning*

- Developing a multimodal AI system to evaluate interview performance using text, audio, and video inputs.
- Implementing RAG pipelines with BERT-based models for contextual analysis, sentiment evaluation, and dynamic question generation.
- Engineered deep learning models (CNN, LSTM) for real-time speech emotion and facial expression recognition.

SKILLS

AI & Cloud: RAG Pipelines, Prompt Engineering, AWS (EC2, S3 - basics)

Languages: Java, JavaScript (ES6+), TypeScript, Python

Frontend/Backend: React.js, Next.js, Node.js, Express.js, Tailwind CSS

Database & Tools: MongoDB, MySQL (basic), Git, GitHub, Postman, Docker, CI/CD (GitHub Actions - basics), Pandas

Concepts: REST APIs, DSA, OOP, DBMS, OS, Linux (basic)

ACHIEVEMENTS & CERTIFICATIONS

- Improved backend response time by 40% and frontend load speed by 25% through performance optimization.
- Secured 3rd place in intra-college hackathon and 2nd place in CSS Battle.
- IBM Python 101 for Data Science – CognitiveClass .